

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO1: Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)

QUESTIONS: 05

Total Marks:100

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I A.sai Rohit (192472144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *A.sai Rohit*

Annexure A

Concept Mapping

1. A college wants to analyze student performance using attendance records, internal marks, assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.
2. A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.
3. An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.
4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.
5. A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

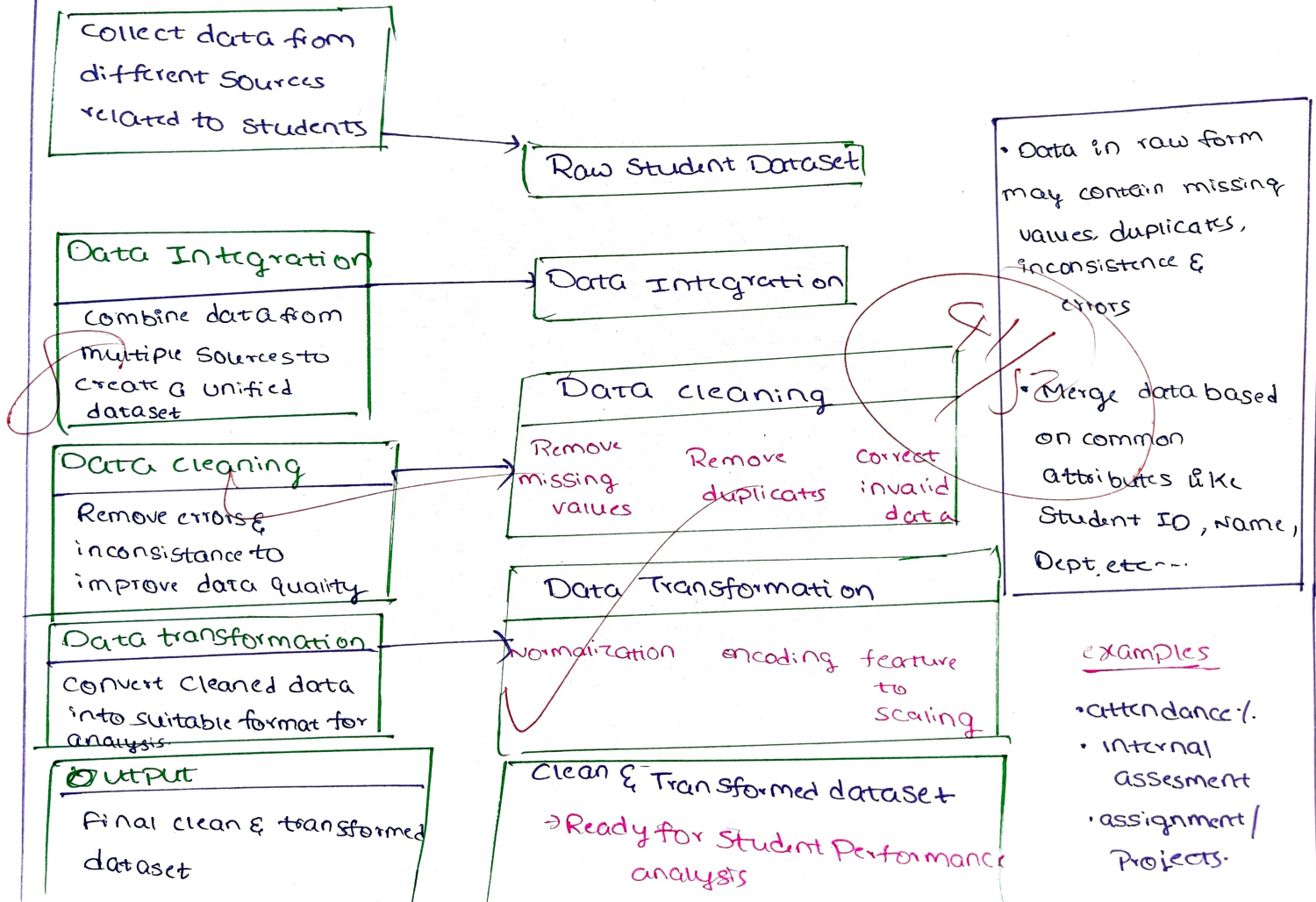
CO1-ASSESSMENT TOOL -2 Concept Map

QUESTIONS : 5

MARKS : 50

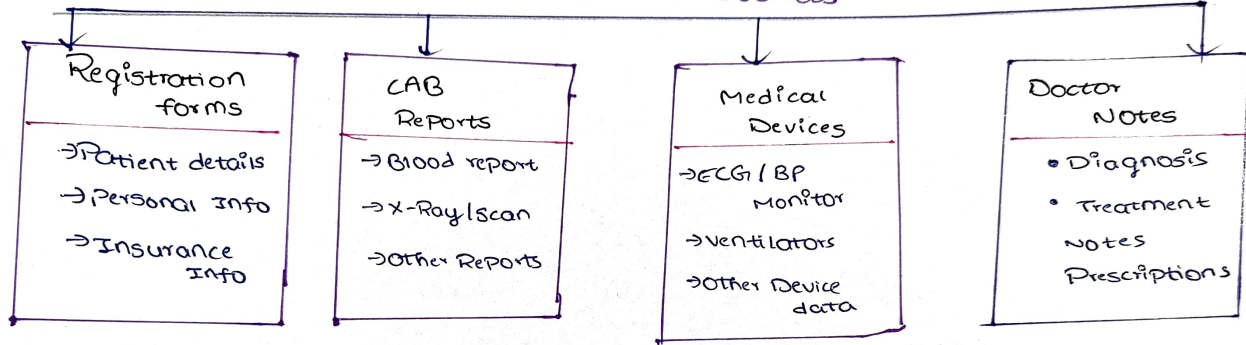
Q No	Concept Identification (25%)	Linkages (25%)	Organisation (20%)	Integration of Literature (20%)	Clarity (10%)	Total Score (100%)
1	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
2	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
3	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
4	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
5	2.5 /2.5	2.5 /2.5	1 /2.0	2 /2.0	1 /1.0	9 /10
Overall Rubrics Value	12.5 /12.5	12.5 /12.5	5 /10	5 /10	5 /10	41 /50
	25 /25	25 /25	10 /20	12 /20	10 /10	82 /100

1. Student Performance Analysis



2. Hospital Data collection and Analysis

Data collection sources



Data collection & Integration (combine data from multiple sources)

Central Databases (All Patient data stored in one place)

Data cleaning

- Removes missing values
- removes duplicate correct errors

Data transformation

- standardization
- formatting data conversion

Data Storage

- Diagnosed data
- Source storage easy retrieval

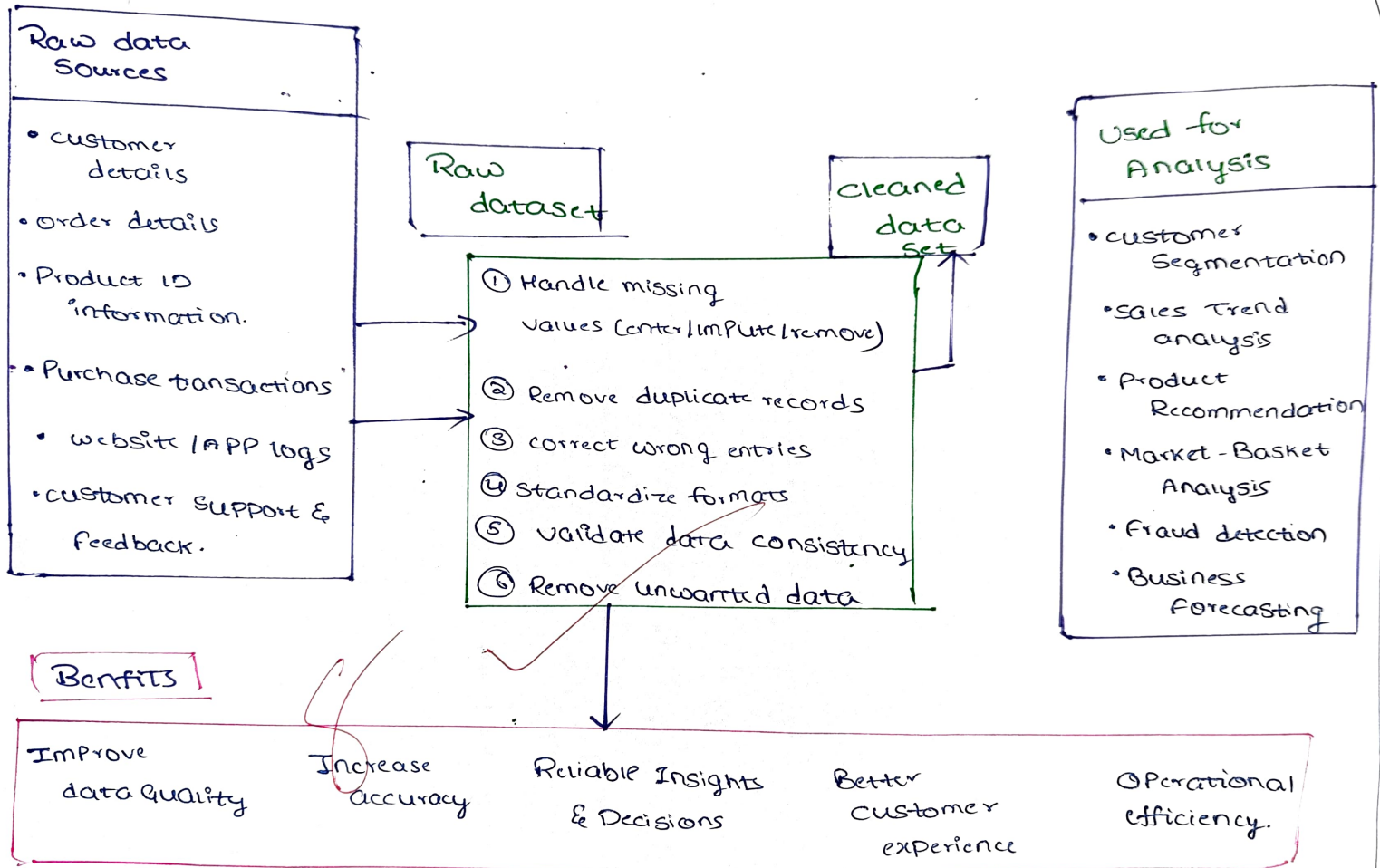
Data Analytics

Disease Prediction Predict risk of diseases using Patient data

Patient Monitoring real-time monitoring of Patient - health

Treatment Planning based on analysis.

3. Online shopping company - Data cleaning

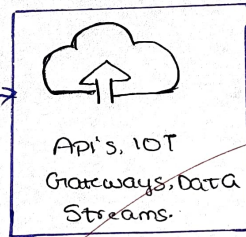


4. Smart city Project

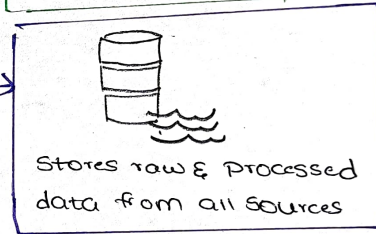
Data Sources

- CCTV cameras
- GPS devices
- Sensors
- Mobile Applications
- Municipal Records
- Social Media & web

Data Ingestion Layer



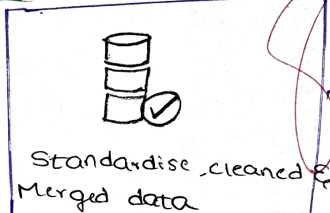
Central data lake/warehouse



Outcome

A smarter, safer & Sustainable city

Integrated Unified dataset



Used for

- Traffic Management
- Public Safety & Surveillance
- Environmental Monitoring
- Smart Governance & planning
- Resource Optimisation
- Citizen Services Improvement

Benefits

- Real-time monitoring
- Better decision making
- efficient resource utilisation

5. Bank Loan Approval Prediction

